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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations
May 2025

AM2001-1 – ADVANCED MACHINE LEARNING

(For AM)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. In k-fold cross-validation, what happens when k is increased?
A) Training time decreases
B) Model bias increases
C) Variance of model performance reduces
D) More data is used for training
2. Which technique is used to systematically search for the best hyperparameters?
A) Cross-validation
B) GridSearchCV
C) Feature engineering
D) Standardization
3. What type of problem does logistic regression solve?
A) Regression
B) Clustering
C) Classification
D) Dimensionality reduction
4. What is the main idea of SVM in classification tasks?
A) Finding a hyperplane that best separates classes
B) Reducing dataset size
C) Creating multiple clusters
D) Increasing dataset features
5. What is the role of a kernel function in SVM?
A) To project data into higher-dimensional space for better classification
B) To reduce the number of features
C) To find missing values
D) To remove outliers
6. Softmax regression is used for:
A) Binary classification
B) Multi-class classification
C) Clustering
D) Regression analysis
7. What is the best way to handle categorical data in a dataset?
A) Ignore categorical columns
B) Convert categorical data into numerical format using encoding techniques
C) Remove all categorical columns
D) Convert all categorical columns into a single numerical value
8. When should you use standardization instead of min-max scaling?
A) When data has outliers
B) When data is normally distributed
C) When using distance-based algorithms like k-NN
D) When scaling should be between 0 and 1
9. What is a common method websites use to block scrapers?
A) CAPTCHA
B) IP blocking
C) User-agent detection
D) All of these
10. What Python library is commonly used to send HTTP requests?
A) requests
B) urllib
C) scrapy
D) flask

11. Which component of Scrapy is responsible for defining how data is extracted?
 A) Pipeline B) Spider
 C) Selector D) Middleware
12. What is a "robots.txt" file used for?
 A) Preventing web crawlers from accessing certain pages B) Blocking users from logging in
 C) Controlling the browser settings D) Storing cookies
13. What is Scrapy?
 A) A web browser B) A Python web scraping framework
 C) A cloud database D) A JavaScript framework
14. Which Python module is used for regular expressions?
 A) regex B) re
 C) BeautifulSoup D) scrapy.regex
15. What is the main purpose of regular expressions in web scraping?
 A) Storing scraped data B) Defining patterns to extract specific information
 C) Automating browser interactions D) Parsing JSON responses
16. What is the purpose of edge detection in images?
 A) To blur the image B) To detect color variations
 C) To find object boundaries D) To improve image contrast
17. Which of the following is NOT a feature extraction technique?
 A) Harris Corner Detector B) SIFT (Scale-Invariant Feature Transform)
 C) K-Means Clustering D) HOG (Histogram of Oriented Gradients)
18. What is the main advantage of using a Gaussian filter in image processing?
 A) It removes high-frequency noise while preserving edges B) It sharpens the image
 C) It enhances image contrast D) It reduces image resolution
19. What does SIFT (Scale-Invariant Feature Transform) do?
 A) Identifies key points in images that remain unchanged under transformations B) Converts an image into grayscale
 C) Smooths an image D) Reduces image size
20. In geotagged image analysis, what type of metadata is commonly used?
 A) File size B) Image resolution
 C) Color depth D) GPS coordinates

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define data cleaning in machine learning. Explain how missing values can be handled using different methods.	8	L*2	1	1
	b) Explain different approaches to handling text and categorical attributes with a code snippet.	8	L2	1	1
2.	a) Define feature scaling, and state the importance of feature scaling in machine learning. Illustrate with a code snippet, any one of the feature selection algorithms.	8	L3	2	2
	b) Delineate the usefulness of pipelines, and discuss with an example.	8	L3	1	2
3.	a) Explain in brief feature selection using Laplacian score and f-score method.	8	L2	2	1
	b) Discuss how feature selection can be done with a hybrid genetic algorithm.	8	L2	2	1

Unit – II

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| 4. | a) | Explain BeautifulSoup and with an example, illustrate how to scrape one objects one at a time. | 8 | L3 | 3 | 2 |
| | b) | Differentiate among, find() and find_all() methods with BeautifulSoup(). | 8 | L2 | 3 | 1 |
| 5. | a) | How regular expressions can be used in scraping? Illustrate with an example. | 8 | L3 | 4 | 2 |
| | b) | Define web crawling. Explain web crawling by traversing a single domain. | 8 | L2 | 3 | 1 |
| 6. | a) | Explain crawling an entire site with an example. | 8 | L2 | 4 | 1 |
| | b) | Define scrapy, and illustrate how to write simple scrapper. | 8 | L2 | 4 | 1 |

Unit – III

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|----|----|---|---|----|---|---|
| 7. | a) | Elaborate on Python Imaging Library (PIL) and Matplotlib library with an example. | 8 | L2 | 5 | 1 |
| | b) | Discuss about image de-noising technique with an example. | 8 | L3 | 5 | 1 |
| 8. | a) | Illustrate with an example, Harris-corner detector with an example. | 8 | L2 | 5 | 1 |
| | b) | Elaborate on the technique Scale Invariant Feature Transform Technique with an example. | 8 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
